

Name: _____

DECISIONS BY DESIGN
Math 260 — Midterm 1 — Wildfires
March 10, 2026

This exam is double-sided with 10 pages and a total of 70 points. You should complete this exam in 75 minutes (assuming no accommodations). There is extra room for scratch work on pages 6 and 7 of the exam.

About this Exam.

- The only technologies you may use are a pen or pencil, the printed pages of this exam, scratch paper, and a calculator (you can use your phone as long as it is in airplane mode and you access nothing other than the calculator).
- Read each question and its instructions carefully.
- Use scratch paper as needed before writing on the exam.
- On the exam pages, show your work neatly and in an organized fashion. Only work on these pages will be graded.
- If there is a box or line for a final answer, please put your final answer in the box or on the line.
- If there is a bubble choice for your answer, please fill in the bubble. Sometimes there is more than one answer to this kind of question; fill in every choice that is correct.
- Sign the honor code below when you finish the exam.

I have neither given nor received unauthorized help. I have not used unauthorized resources. I have adhered to the time limit (within 5 minutes).

Signed, _____

Turn in your Midterm and Exam Packet; make sure your name is on both!

Part I. Mathematical Modeling

The exam packet provides a Mathematical Modeling Brief describing an intervention to solve a problem (Step 6 - Reporting the Results). For each of the other five steps in the M3 Mathematical Modeling Process, describe how the step is playing out or indicate that you don't have enough information from the report to do so. Underline the parts of the text you are using to inform your answers.

1. (5 points) (Defining the Problem Statement) What problem are researchers trying to solve?

2. (5 points) (Making Assumptions) What assumptions are they making to reduce complexity?

3. (5 points) (Defining Variables) What variables have the researchers identified as being of primary importance to the problem?

4. (5 points) (Getting a Solution) What type of model and/or tools are the researchers using to solve the problem?

5. (5 points) (Analysis and Model Assessment) What are the researchers doing to understand how well the model works to solve the problem?

Part II. Apportionment

Welcome to Flatland! In Flatland, the Unified Republic of Graphpapyra (URG, or “urg” as they call themselves; pop. 27000) is a country that has a representative system of government. URG has four states: Cartesia (pop. 10290), Gridlandia (pop. 1900), the Commonwealth of Squares (pop. 6100), and North Boxes (pop. 8710).¹

Although life is fairly simple in Flatland and URG, residents still face problems we recognize from our lives, like the problem of allocating representatives for these four states based on their populations. Their legislative body, the URG Assembly (URGA or “urg-guh”) has 100 seats.

6. (3 points) Based on the information above, the standard divisor is _____.
7. (5 points) Complete Table 1 in the Exam Packet to apportion the seats by Hamilton’s method (every gets their lower quota; biggest decimal remainders are awarded the extra seats).
8. (2 points) Webster’s method uses traditional rounding ($n \geq 0.5$ rounds to $n + 1$, $n < 0.5$ rounds to n) with a divisor that ensures the rounded numbers sum to 100. Does the standard divisor lead to a valid Webster’s apportionment? ☐ Yes ☐ No
9. (5 points) South Boxes sees that life in URG has been steadily improving and decides to return to the Republic with its 2845 residents. The URGA adds 10 news seats to accommodate their return. Compute the new apportionment in Table 2.
10. (5 points) One of the Squares assembly members is unhappy and tries to convince URGA to use Webster’s method with a modified divisor of 271. Compute the proposed apportionment in Table 3, then explain why this would threaten the fragile reunification.



¹South Boxes (pop. 2845) seceded following an uncivil war.

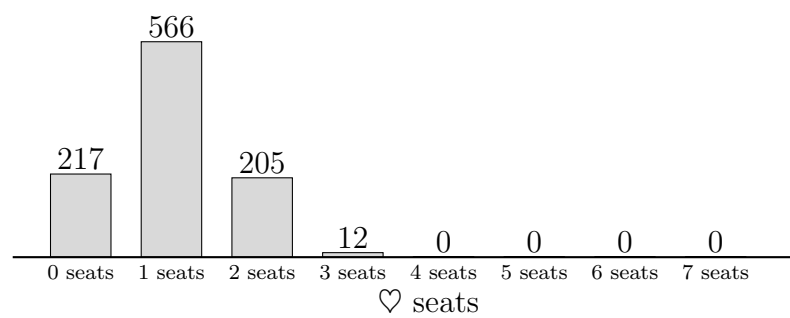
Part III. Districting Questions

In the Exam Packet, you will find grids relating to this question.

It’s time for redistricting in URG! By precinct, voters prefer the Clubs Party or the Hearts Party.²

In the MidSouth, there is a minority voting bloc indicated with large hearts. URG and Gridlandia laws dictate that these voters must have an opportunity district where their views are in the majority.

To ensure fairness, Gridlandia has created a task force to analyze proposed district maps. The task force runs an MCMC analysis of district plans given the distribution of voter preferences in prior elections which returns the following distribution:



11. (5 points) Compute the number of seats each party wins in the district plans on the last page. *You will complete Plan 1' in a later question; compute the numbers of seats when your plan is complete.*

Plan	1	2	3	4	1'
♣ seats					
♡ seats					

12. (1 point) Based on the distribution above, the task force will reject one of the plans on the next page. Which one? ☐ Plan 1 ☐ Plan 2 ☐ Plan 3 ☐ Plan 4
13. (1 point) Based on the law protecting the right of representation for minority voting blocs, which plans will the task force reject? ☐ Plan 1 ☐ Plan 2 ☐ Plan 3 ☐ Plan 4
14. (3 points) Plan 1 can be modified to create an opportunity district by modifying districts 5 and 7. Do so on the grid labeled Plan 1' on the next page.

²Conveniently, the populated region of Gridlandia is a 7×7 grid.

Define $p(D)$ to be the perimeter of a district D , $a(D)$ to be the area of a district D , and $MP(D)$ is the minimal convex polygon containing D .

15. (5 points) Compute *and interpret* the Polsby-Popper score for $D_{3,3}$ (Plan 3 District 3),

$$\text{polpop}(D_{3,3}) = \frac{4\pi a(D_{3,3})}{p(D_{3,3})^2}.$$

16. (5 points) Compute *and interpret* the convex polygon score for $D_{1,6}$ (Plan 1 District 6),

$$\text{convex}(D_{1,6}) = \frac{a(D_{1,6})}{a(MP(D_{1,6}))}.$$

17. (5 points) Compute the Efficiency Gap for Plan 2 by completing the table below, where

$$\text{EG} = \frac{\text{wasted}(\clubsuit) - \text{wasted}(\heartsuit)}{49}.$$

$$\text{EG} = \underline{\hspace{2cm}}$$

A wasted vote in a district is a vote for the losing candidate or a vote for the winning candidate that was unneeded.

District	 votes	 votes	Winner	 wasted votes	 wasted votes
1	4	3		0	3
2					
3					
4					
5					
6					
7					
totals:					

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(Scratch Work Page)

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Part I. Mathematical Modeling Brief

MATHEMATICAL MOMENTS



Fighting Fires

In many places, fire seasons keep getting longer with larger and ever more destructive wildfires. Teams of mathematicians, computer scientists, meteorologists, and firefighters are working to reduce the number of large fires before they happen and to contain those that do occur. Reducing the number and size involves employing wiser land-use strategies to decrease the amount of fuel available, while containing fires that are already burning entails collecting and processing vast amounts of data quickly so firefighters can act to save lives and property. Frequently-updated information about factors such as wind speed, moisture levels, and terrain is fed into numerical models of fire spread (which are also updated based on their performance for that fire) that use computational techniques to make rapid predictions about the areas to be affected. Firefighters on the scene use those predictions to guide their strategies for deploying crews and conducting evacuations.

Climate change affects the frequency of various weather conditions that determine the likelihood and growth potential of wildfires, and fires create their own weather, making them even more difficult to predict. In some cases, the fire can react with turbulent air and create a fire tornado, one of which in California had winds estimated at over 150 mph and temperatures around 1500°C. Mathematicians and other researchers are working in the lab and on the ground to understand “firenadoes” and related properties, such as flame structure (flames’ highs and lows are not totally random), the *heat release rate* (which can cause objects 30 feet away from a fire to ignite), and how fire spreads by both convection and radiation. The challenge is to find simple physical and mathematical ways to describe complex fire phenomena so as to improve operational uses of models for fire managers.



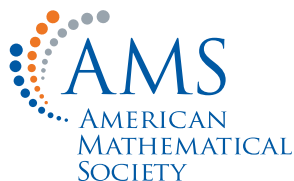
©Getty Images.

For More Information: “Firemap: A Dynamic Data-Driven Predictive Wildfire Modeling and Visualization Environment,” D. Crawl, J. Block, K. Lin, and I. Altintas, *Procedia Computer Science*, 2017, pp. 2230-2239.

Listen Up!



MM/154



The **Mathematical Moments** program promotes appreciation and understanding of the role mathematics plays in science, nature, technology, and human culture.

www.ams.org/mathmoments

Part II. Apportionment Tables

State	Pop.	Quota	Upper Quota	Lower Quota	Hamilton Seats
Gridlandia	1900				
Squares	6100				
N. Boxes	8710				
Cartesia	10290				
Totals					

Table 1: UGRA Four State Apportionment.

State	Pop.	Quota	Upper Quota	Lower Quota	Hamilton Seats
Gridlandia	1900				
Squares	6100				
N. Boxes	8710				
Cartesia	10290				
S. Boxes	2845				
Totals					

Table 2: UGRA Five State Apportionment.

State	Pop.	Modified Quota	Webster Seats
Gridlandia	1900		
Squares	6100		
N. Boxes	8710		
Cartesia	10290		
S. Boxes	2845		
Totals			

Table 3: URG Five State Apportionment with modified divisor.

Part III. Districting Grids

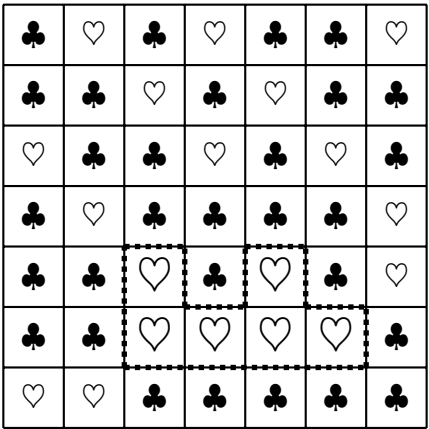
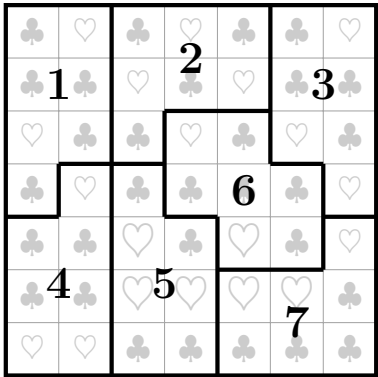
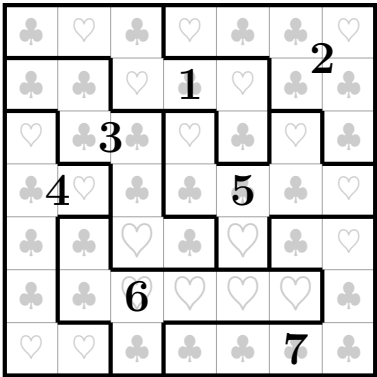


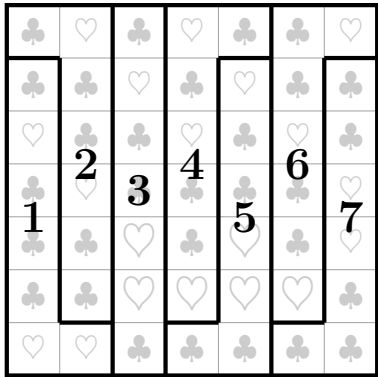
Figure 1: Gridlandia Precinct Preferences.
Large Hearts form a minority voting bloc.



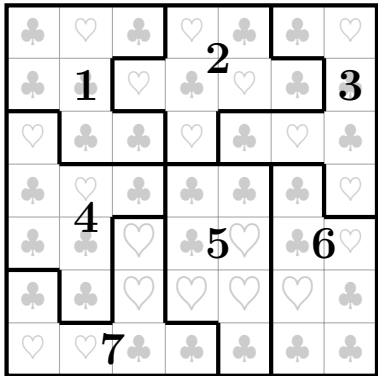
(a) Plan 1.



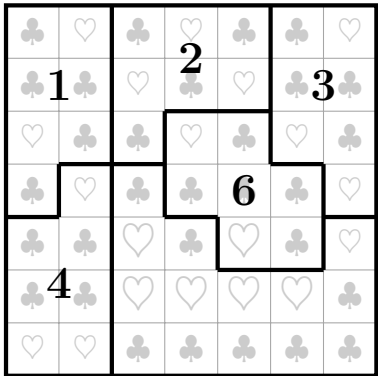
(b) Plan 2.



(c) Plan 3.



(d) Plan 4.



(e) Plan 1'. (DIY)